

Giovanni Bordiga

Curriculum Vitae

Research interests

Mechanical intelligence · Inverse design · Differentiable physics · Optimization · Mechanical metamaterials · Material instabilities · Wave propagation · Homogenization theory

Education

- Nov. 2016 – **PhD in Solid and Structural Mechanics**,
Apr. 2020 *Doctoral School of Civil, Environmental, and Mechanical Engineering*,
University of Trento (Italy),
Curriculum: Modeling and Simulation
- PhD Thesis: “Homogenization of periodic lattice materials for wave propagation, localization, and bifurcation”,
Advisors: Andrea Piccolroaz and Davide Bigoni,
Evaluation Committee: Basile Audoly and Pedro M. Reis,
Final grade: *Cum Laude*.
- Sep. 2014 – **MSc in Civil Engineering**,
Oct. 2016 *Department of Civil, Environmental, and Mechanical Engineering*,
University of Trento (Italy),
Curriculum: Structures
- MSc Thesis: “Micromechanical modeling of masonry” (in Italian),
Advisor: Davide Bigoni,
Final grade: *110/110 Cum Laude*.
- Sep. 2011 – **BSc in Civil Engineering**,
Sep. 2014 *Department of Civil, Environmental, and Mechanical Engineering*,
University of Trento (Italy)
- BSc Thesis: “Non-uniform torsion of thin-walled open cross-section beams” (in Italian),
Advisor: Massimiliano Gei,
Final grade: *110/110 Cum Laude*.

Research experience

- Sep. 2021 – **Postdoctoral Fellow**,
present *Harvard School of Engineering and Applied Sciences*,
Harvard University (US),
Advisors: Katia Bertoldi
- Development of a differentiable simulation framework for automating the design of flexible mechanical metamaterials with target dynamic responses.
 - Inverse design of flexible mechanical metamaterials for shape morphing, nonlinear wave guiding, energy focusing, shock mitigation, and cloaking of nonlinear waves.
- May 2020 – **Postdoctoral Researcher**,
Aug. 2021 *Department of Civil, Environmental, and Mechanical Engineering*,
University of Trento (Italy),
Advisors: Andrea Piccolroaz and Davide Bigoni

- Homogenization of elastic lattices prestressed by circulatory (follower) forces demonstrating the existence of hypoelastic continuum materials displaying flutter instability at the macroscale.
- Analytical and numerical investigation of tensile buckling in periodic lattice materials endowed with sliding constraints.

Nov. 2016 – **PhD Student in Solid and Structural Mechanics,**
 Apr. 2020 *Department of Civil, Environmental, and Mechanical Engineering,*
University of Trento (Italy),
 Advisors: Andrea Piccolroaz and Davide Bigoni

- Analytical and numerical investigation of the free and forced dynamic response of Rayleigh-beam lattices leading to sharp frequency-dependent wave localization.
- Modeling and tuning of the prestress state in a lattice domain to realize dynamic interfaces designed for total reflection, negative refraction, and wave channeling.
- Development of static and dynamic homogenization methods for beam lattices prestressed by conservative and non-conservative positional forces.
- Analytical and numerical analysis of macroscopic and microscopic bifurcations in prestressed anisotropic lattices and their connection to shear bands formation and short-wavelength instabilities.

Entrepreneurial experience

Apr., 2024 – **Founder & Chief Technology Officer (CTO),**
 present *SoftPulse, Inc.,* Cambridge, MA (US),
 Mission: Develop a passive implant for lasting blood pressure control. A drug-free solution for millions with hypertension.
 Team: Adel Djellouli, Giovanni Bordiga, and Katia Bertoldi.
 Funding: [\\$160K from the Harvard Grid Accelerator.](#)

Publications

Preprints

- [2] A. A. Watkins, **G. Bordiga**, V. Tournat, K. Bertoldi, *Wave-based reading of mechanical memory in multistable mass-in-mass metamaterials*. 2026. DOI: [10.48550/arXiv.2609.07889](#). Pre-published.
- [1] N. Singh, A. A. Watkins, **G. Bordiga**, V. Tournat, K. Bertoldi, Z. Rocklin, *Conformal Elastodynamics in 2D Dilational Metamaterials*. 2026. DOI: [10.48550/arXiv.2604.16637](#). Pre-published.

Peer-reviewed journals

- [11] A. A. Watkins, **G. Bordiga**, M. Mu, V. Tournat, K. Bertoldi, “Arbitrary mechanical memory encoding via nonlinear waves in bistable metamaterials”. *Phys. Rev. Lett.* 137.9 (2026), p. 097201. DOI: [10.1103/pkk5-dykb](#).
- [10] **G. Bordiga**, J.-G. Argaud, A. A. Watkins, V. Tournat, K. Bertoldi, “Nonlinear mechanical metamaterial cloaks”. *Adv. Funct. Mater.* (2025), e22895. DOI: [10.1002/adfm.202522895](#).
- [9] C. Perez-Garcia, R. Zaera, J. Aranda-Ruiz, **G. Bordiga**, G. Risso, M. L. Lopez-Donaire, K. Bertoldi, D. Garcia-Gonzalez, “Reprogrammable mechanical metamaterials via passive and active magnetic interactions”. *Adv. Mater.* (2025), p. 2412353. DOI: [10.1002/adma.202412353](#).
- [8] A. S. Meeussen, **G. Bordiga**, A. X. Chang, B. Spoettling, K. P. Becker, L. Mahadevan, K. Bertoldi, “Textile hinges enable extreme properties of kirigami metamaterials”. *Adv. Funct. Mater.* (2024), p. 2415986. DOI: [10.1002/adfm.202415986](#).
- [7] **G. Bordiga**, E. Medina, S. Jafarzadeh, C. Bösch, R. P. Adams, V. Tournat, K. Bertoldi, “Automated discovery of reprogrammable nonlinear dynamic metamaterials”. *Nat. Mater.* 23.11 (2024), pp. 1486–1494. DOI: [10.1038/s41563-024-02008-6](#).

- [6] A. Kotikian, A. A. Watkins, **G. Bordiga**, A. Spielberg, Z. S. Davidson, K. Bertoldi, J. A. Lewis, “Liquid crystal elastomer lattices with thermally programmable deformation via multi-material 3D printing”. *Adv. Mater.* 36.34 (2024), p. 2310743. DOI: [10.1002/adma.202310743](https://doi.org/10.1002/adma.202310743).
- [5] **G. Bordiga**, D. Bigoni, A. Piccolroaz, “Tensile material instabilities in elastic beam lattices lead to a bounded stability domain”. *Philos. Trans. R. Soc. Math. Phys. Eng. Sci.* 380.2231 (2022), p. 20210388. DOI: [10.1098/rsta.2021.0388](https://doi.org/10.1098/rsta.2021.0388). (Cover article).
- [4] **G. Bordiga**, A. Piccolroaz, D. Bigoni, “A way to hypo-elastic artificial materials without a strain potential and displaying flutter instability”. *J. Mech. Phys. Solids* 158 (2022), p. 104665. DOI: [10.1016/j.jmps.2021.104665](https://doi.org/10.1016/j.jmps.2021.104665).
- [3] **G. Bordiga**, L. Cabras, A. Piccolroaz, D. Bigoni, “Dynamics of prestressed elastic lattices: Homogenization, instabilities, and strain localization”. *J. Mech. Phys. Solids* 146 (2021), p. 104198. DOI: [10.1016/j.jmps.2020.104198](https://doi.org/10.1016/j.jmps.2020.104198).
- [2] **G. Bordiga**, L. Cabras, D. Bigoni, A. Piccolroaz, “Free and forced wave propagation in a Rayleigh-beam grid: Flat bands, Dirac cones, and vibration localization vs isotropization”. *Int. J. Solids Struct.* 161 (2019), pp. 64–81. DOI: [10.1016/j.ijsolstr.2018.11.007](https://doi.org/10.1016/j.ijsolstr.2018.11.007).
- [1] **G. Bordiga**, L. Cabras, A. Piccolroaz, D. Bigoni, “Prestress tuning of negative refraction and wave channeling from flexural sources”. *Appl. Phys. Lett.* 114.4 (2019), p. 041901. DOI: [10.1063/1.5084258](https://doi.org/10.1063/1.5084258). (Editor’s Pick).

In conference proceedings

- [3] **G. Bordiga**. “In-plane Floquet-Bloch waves in elastic grids and prestress tuning of structured interfaces”. *RAMSS2019 - Recent Advances in Mechanics of Solids and Structures*. Trento, 2019. URL: <http://rgdoi.net/10.13140/RG.2.2.15758.82244>.
- [2] **G. Bordiga**. “Free and forced wave propagation in a Rayleigh-beam grid: flat bands, Dirac cones, and vibration localization vs isotropization”. *ESMC2018 - 10th European Solid Mechanics Conference*. Bologna, 2018. URL: <http://rgdoi.net/10.13140/RG.2.2.24679.09126>.
- [1] **G. Bordiga**. “Micromechanical modelling of masonry”. *CERMODEL2017*. Trento, 2017.

Software

- [3] **G. Bordiga**. *DifFlexMM: Differentiable Flexible Mechanical Metamaterials*. Bertoldi Group, 2024. URL: <https://github.com/bertoldi-collab/DifFlexMM>.
- [2] **G. Bordiga**. *Simulation and Design of Shape-Morphing LCE Lattices*. Bertoldi Group, 2024. URL: <https://github.com/bertoldi-collab/morphing-lattices>.
- [1] **G. Bordiga**. *A humble image tracking code*. Bertoldi Group, 2023. URL: <https://github.com/bertoldi-collab/tracking-markers>.

Conference presentations

- Mar. 16–20, **APS2026 Global Physics Summit**, Denver, CO (US),
2026 *Tuning nonlinear mechanical metamaterials for fully-analog sensing and robotic control*,
G. Bordiga, C. Bösch, C. McCann, M. Yuen, J. Wilt, S. Jafarzadeh, K. Bertoldi.
- Mar. 16–21, **APS2025 Global Physics Summit**, Anaheim, CA (US),
2025 *Nonlinear mechanical metamaterial cloaks*,
G. Bordiga, J. Argaud, A. A. Watkins, V. Tournat, K. Bertoldi.
- Sep. 9–11, **SMASIS2024**, Atlanta, GA (US),
2024 *Automated design of flexible mechanical metamaterials with reprogrammable wave functionalities*,
G. Bordiga, E. Medina, S. Jafarzadeh, C. Bösch, V. Tournat, K. Bertoldi.
- Sep. 9–11, **SMASIS2024**, Atlanta, GA (US),
2024 *Mechanical metamaterial ‘brain’ for fully analog control of a mobile robot*,
C. Bösch, **G. Bordiga**, C. McCann, S. Jafarzadeh, J. Wilt, M. Yuen, Y. Jin, A. Fichtner, K. Bertoldi.

- Jul. 3–4, 2024 **DynaMetaFlex**, Workshop on “Nonlinear dynamics of flexible mechanical metamaterials”, Laboratoire d’Acoustique de l’Université du Mans, Institut d’Acoustique – Graduate School, CNRS, Le Mans Université, Le Mans, (France),
Teaching reprogrammable dynamics to material structures,
G. Bordiga, E. Medina, S. Jafarzadeh, C. Bösch, V. Tournat, K. Bertoldi.
- Oct. 8–11, 2023 **SES2023**, Minneapolis, MN (US),
Inverse-design of nonlinear mechanical metamaterial cloaks,
G. Bordiga, J. Argaud, V. Tournat, K. Bertoldi.
- Mar. 5–10, 2023 **APS2023**, Las Vegas, NV (US),
Manipulating energy flows with non-periodic mechanical metamaterials,
G. Bordiga, E. Medina, V. Tournat, K. Bertoldi.
- Oct. 16–19, 2022 **SES2022**, College Station, TX (US),
Non-periodic design discovery for optimal dynamic responses in flexible mechanical metamaterials,
G. Bordiga, E. Medina, V. Tournat, K. Bertoldi.
- Jul. 4–8, 2022 **ESMC2022**, 11th European Solid Mechanics Conference, Galway (Ireland),
Non-hyper-elastic materials from follower prestress states reveal flutter instability in elastic lattices (Invited talk),
G. Bordiga, A. Piccolroaz, D. Bigoni.
- Mar. 14–18, 2022 **APS2022**, Chicago, IL (US),
Architected metamaterials for routing nonlinear mechanical pulses,
G. Bordiga, E. Medina, V. Tournat, K. Bertoldi.

Invited talks & Seminars

- Jun. 22–25, 2026 **NSF ID4 Workshop**, [Accelerating Quantum Materials Discovery Workshop](#), Colorado School of Mines, Golden, CO (US),
Nonlinear mechanical metamaterial cloaks,
G. Bordiga, J. Argaud, A. A. Watkins, V. Tournat, K. Bertoldi.
- Mar. 31, 2025 **Seminar**, [Center for Fluid Mechanics Seminar](#), Brown University, Providence, RI (US),
Automating the discovery of nonlinear architected materials,
G. Bordiga.
- Jul. 25, 2024 **Seminar**, Seismology and Wave Physics Lab, ETH, Zürich (Switzerland),
Teaching reprogrammable dynamics to material structures,
G. Bordiga.
- Jun. 13, 2024 **Seminar**, Laboratory for Intelligent Probabilistic Systems, Princeton University, Princeton, NJ (US),
Teaching reprogrammable dynamics to material structures,
G. Bordiga.
- Jul. 4–8, 2022 **ESMC2022**, 11th European Solid Mechanics Conference, Galway (Ireland),
Non-hyper-elastic materials from follower prestress states reveal flutter instability in elastic lattices,
G. Bordiga, A. Piccolroaz, D. Bigoni.

Media coverage

About my research

- Sep. 15, 2025 **Colorado School of Mines**, [The Institute for Data Driven Dynamical Design \(ID4\)](#),
[ID4 Research Highlights on Dynamic Metamaterials](#)
- May 7, 2025 **Azo Materials**,
[Novel Mechanical Metamaterials with Magnetically Controlled Properties](#)
- May 6, 2025 **Tech Xplore**,
[Structurally reprogrammable magnetic metamaterials hold promise for biomedicine, soft robotics](#)
- Oct. 17, 2024 **Nature Materials News and Views**, Gary P. T. Choi,
[Designing flexible mechanical metamaterials with complex functionalities](#)

- Oct. 31, 2024 **Communication CNRS Ingénierie,**
La conception de métamatériaux guidée par des outils de l'IA
[About SoftPulse, Inc.](#)
- Jul. 8, 2025 **The Harvard Gazette,** Kirsten Mabry, Harvard Office of Technology Development,
3 tech solutions to societal needs will get help moving to market

Mentoring

At Bertoldi Lab, Harvard University

- Sep., 2025 – **Yanzi Wang**, Undergraduate student, **Tsinghua University (China)**,
present Project: Mechanical metamaterials for underactuated dynamic path generation.
- Sep., 2024 – **Louis-Justin Tallot**, PhD student, **Harvard University (US)**,
present Project: Differentiable origami simulation; Inverse-design of magneto-mechanical metamaterials.
- Sep., 2024 – **Stijn de Bruin**, Master student, **TU Delft (Netherlands)**,
Dec., 2025 Project: Encoding dynamic bifurcations in mechanical metamaterials.
- Sep., 2024 – **Antoine Fondeur**, Master student, **MINES Paris – PSL (France)**,
Jan., 2025 Project: Inverse design of magneto-mechanical metamaterials for reprogrammable static and dynamic responses.
- Sep., 2024 – **Jian Zhimo**, PhD student, **Tsinghua University (China)**,
Jan., 2025 Project: Mechanical metamaterials for sensing and control.
- Sep., 2023 – **Tom Vreugdenhil**, Master student, **TU Delft (Netherlands)**,
Sep., 2024 Thesis: Dynamic metamaterials for reprogrammable underactuated robotics.
- Sep., 2023 – **Carlos Pérez García**, PhD student, **University Carlos III of Madrid (Spain)**,
Dec., 2024 Project: Magneto-mechanical metamaterials for reprogrammable stress-strain response and energy absorption.
- Feb.–Sep., **Jean-Gabriel Argaud**, Master student, **MINES Paris – PSL (France)**,
2023 Project: Cloaking in nonlinear mechanical metamaterials.
- Jul.–Dec., **Cyрил Bösch**, PhD student, **ETH (Switzerland)**,
2022 Project: Mechanical metamaterial 'brain' enables fully analog robotic control.
- Sep., **Audrey Watkins**, PhD student, **Harvard University (US)**,
2022–present Project: Nonlinear waves in multistable metamaterials for mechanical computing.
- Feb.–Sep., **Ben Spoettling**, Master student, **ETH (Switzerland)**,
2022 Thesis: Automated design and discovery of shape-morphing metamaterials.

Teaching experience

- 2018–2020 **Teaching assistant,**
Department of Civil, Environmental, and Mechanical Engineering,
University of Trento (Italy),
Course: *Structural Dynamics*
- 40+ students.
 - Material: Created interactive Mathematica notebooks to teach the dynamics of single and multi-degree-of-freedom systems, modal analysis, free and forced vibrations of elastic beams.
 - Graded: oral exams.
- 2019–2020 **Lecturer and tutor,**
Department of Civil, Environmental, and Mechanical Engineering,
University of Trento (Italy),
Course: *Solids and Structural Mechanics*
- 80+ students, 20 hours of lectures, and 5 hours of tutoring activities.
 - Taught: De Saint Venant theory and exercises, Principle of Virtual Work for statically indeterminate structures, structural symmetries, frame structures.

- Graded: written and oral exams.

2018–2019 **Lecturer and tutor,**
Department of Civil, Environmental, and Mechanical Engineering,
University of Trento (Italy),
 Course: *Solids and Structural Mechanics*

- 80+ students, 20 hours of lectures, and 20 hours of tutoring activities.
- Taught: De Saint Venant theory and exercises, Principle of Virtual Work for statically indeterminate structures, Mohr's circles, elastic beam theory, frame structures.
- Graded: written and oral exams.

2017–2018 **Lecturer and tutor,**
Department of Civil, Environmental, and Mechanical Engineering,
University of Trento (Italy),
 Course: *Solids and Structural Mechanics*

- 80+ students, 30 hours of lectures, and 30 hours of tutoring activities.
- Taught: De Saint Venant theory and exercises, Mohr's circles, failure and yield criteria (Von Mises, Tresca, Mohr-Coulomb, Rankine, Drucker-Prager), Principle of Virtual Work for statically indeterminate structures, elastic beam theory.
- Graded: written and oral exams.

Peer reviews

- [Journal of the Mechanics and Physics of Solids](#)
- [Science Advances](#)
- [Advanced Engineering Materials](#)
- [International Journal of Solids and Structures](#)
- [European Journal of Mechanics - A/Solids](#)
- [Proceedings of the Royal Society A: Mathematical, Physical and Engineering Sciences](#)
- [Computer Methods in Applied Mechanics and Engineering](#)
- [Journal of Elasticity](#)
- [Mechanics of Materials](#)
- [Journal of Sound and Vibration](#)
- [Meccanica](#)
- [Journal of Mechanics of Materials and Structures](#)
- [ACS Photonics](#)

Events & Conferences service

- Mar. 15–20, 2026 **APS2026 Global Physics Summit**, Denver, CO (US),
Session: Functionality Through Nonlinearity
 Served as: **Session chair.**
- Mar. 15, 2026 **Squishy Science Sunday at APS2026 Global Physics Summit**, Denver, CO (US),
Activity: Marbelous Marbling
 Served as: **Volunteer.**
- Sep. 9–11, 2024 **SMASIS2024**, Atlanta, GA (US),
Special Symposium: Embodying Physical Computing and Mechano-Intelligence
 Served as: **Session chair.**
- Jul. 22, 2024 **MRSEC Workshop**, Harvard SEAS, Cambridge, MA (US),
Git + GitHub Workshop: Version control for scientists
 Served as: **Organizer and presenter.**

- Jul. 29, 2022 **MRSEC Workshop**, Harvard SEAS, Cambridge, MA (US),
Git + GitHub Workshop: Version control for scientists
 Served as: **Organizer**.
- Sep. 27, 2019 **European Researchers' night**, Trento (Italy)
 Served as: **Co-organizer and presenter** for the Solid and Structural Mechanics exhibition stand.
- Jun. 6–7, 2019 **RAMSS2019 — Recent Advances in Mechanics of Solids and Structures**, Trento (Italy)
 Served as: **Conference assistant**.
- Sep. 28, 2018 **European Researchers' night**, Trento (Italy)
 Served as: **Co-organizer and presenter** for the Solid and Structural Mechanics exhibition stand.
- Jul. 2–6, 2018 **ESMC2018 — 10th European Solid Mechanics Conference**, Bologna (Italy)
 Served as: **Co-organizer** for conference program.
- Sep. 29, 2017 **European Researchers' night**, Trento (Italy)
 Served as: **Co-organizer and presenter** for the Solid and Structural Mechanics exhibition stand.

Awards & Honors

- 2024 [\\$160K from the Harvard Grid Accelerator](#) to de-risk and develop [SoftPulse](#) hypertension implant towards animal trials.
- 2022 Best PhD Thesis Award for Cycle XXXII, University of Trento (Italy)
- 2017 Graduation Award for Merit, University of Trento (Italy)
- 2016 University Scholarship for Merit, Italian government, INPS
- 2015 University Scholarship for Merit, Italian government, INPS
- 2015 Graduation Award for Merit, University of Trento (Italy)
- 2014 University Scholarship for Merit, Italian government, INPS
- 2013 University Scholarship for Merit, Italian government, INPS
- 2012 University Scholarship for Merit, Italian government, INPS

Community activities

- Jun.–Aug., 2025 **Beach volleyball tournament**, *Harvard Rhino League*, Cambridge, MA (US)
 Role: Team captain for the Bertoldi Lab (2nd place).
- Jun.–Aug., 2024 **Beach volleyball tournament**, *Harvard Rhino League*, Cambridge, MA (US)
 Role: Team captain for the Bertoldi Lab (Championship winner).
- Jun.–Aug., 2023 **Beach volleyball tournament**, *Harvard Rhino League*, Cambridge, MA (US)
 Role: Team captain for the Bertoldi Lab (Championship winner).

Skills

Programming

Scientific computing	Wolfram Language	<i>Highly specialized</i>	Python	<i>Highly specialized</i>
	MATLAB	<i>Highly specialized</i>	R	<i>Intermediate</i>
Markup	L ^A T _E X	<i>Highly specialized</i>	Markdown	<i>Advanced</i>
	HTML/CSS	<i>Basic</i>		
Others	Java	<i>Advanced</i>	C/C++	<i>Intermediate</i>
	Javascript	<i>Advanced</i>	Bash	<i>Intermediate</i>
	Android	<i>Basic</i>	SQLite	<i>Basic</i>

Application software

Modeling & Simulation	Mathematica	<i>Highly specialized</i>	COMSOL	<i>Advanced</i>
	ABAQUS	<i>Advanced</i>	SAP2000	<i>Intermediate</i>
CAD	AutoCAD	<i>Highly specialized</i>	SketchUp	<i>Advanced</i>

IDE	VSCode	<i>Highly specialized</i>	IntelliJ IDEA	<i>Advanced</i>
	Android Studio	<i>Intermediate</i>		
Graphics	Inkscape	<i>Highly specialized</i>	GIMP	<i>Intermediate</i>
Others	Git	<i>Highly specialized</i>	GitHub	<i>Highly specialized</i>
	Zotero	<i>Highly specialized</i>		
Operating systems				
	Linux	<i>Highly specialized</i>	Windows	<i>Advanced</i>
	Android	<i>Advanced</i>	MacOS	<i>Intermediate</i>

Languages

- **Italian:** Native language
- **Chinese:** Duolingo (A1)
- **English:** Professional proficiency